Pending the full implementation of the MDI Rules, the market for market data is serviced by the two exclusive SIPs and exchange proprietary feeds. The two exclusive SIPs collect trade, quote, and regulatory data from the 16 exchanges and three trade reporting facilities,[1158] consolidate the data, determine an NBBO, and disseminate those data directly to users or through vendors and broker-dealers. The exclusive SIPs can also collect information from the alternative display facility ("ADF") operated by FINRA, though no one currently uses the ADF to display quotes. Upon full implementation of the MDI Rules, the exclusive SIPs will be retired, and an unknown number of competing consolidators will take over the collection, consolidation, estimation, and dissemination of these data.[1159] The volume of data to be processed through these competing consolidators will be greater than that currently processed through exclusive SIPs, but competing consolidators will have flexibility to design data products tailored to different user types. In addition to the exclusive SIPs, the exchanges also disseminate market data to paying subscribers via proprietary data feeds. Some of these proprietary data feeds provide more data than the exclusive SIPs and are provided at a lower latency; however, the proprietary feeds are limited to individual exchanges while the SIPs contain consolidated data across all exchanges and also contain all off-exchange trades.[1160] Following the transition to a competing consolidator

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[1158] Trade Reporting Facilities (TRFs) are facilities through which FINRA members report off-exchange transactions in NMS stocks, as defined in SEC Rule 600(b)(47) of Regulation NMS.

[1159] While the Commission is uncertain about the number of competing consolidators that will enter the market when exclusive the SIPs are retired, the Commission believes that the most likely outcome is three or more competing consolidators with at least one competing consolidator that is not affiliated with one of the exchanges currently operating the exclusive SIPs or an exchange that has sufficient proprietary data revenue that would create conflicting profit incentives. See MDI Adopting Release, supra note 10, at 18768-72 for further discussion on the number of competing consolidators that may enter the market.

[1160] See supra note 862 and infra note 1780 and associated text for a further discussion on the nature of proprietary data feeds.

model for market data, the Commission expects total fees for market data are likely to decline.[1161]

Broker-dealers typically route their own orders or their customers' orders for execution to trading venues. There were 3,494 registered broker-dealers as of Q2 2023.[1162] A portion of these broker-dealers focus their business on individual and/or institutional investors in the market for NMS stocks. According to CAT data, as of the end of 2022, there were approximately 1,006 registered broker-dealers that originated NMS stock orders on behalf of individual investors and approximately 837 broker-dealers that originated NMS stocks orders on behalf of institutional investors.[1163] Institutional investor orders are typically "not held" orders, which provides the broker-dealer with more time and price discretion to execute the order or to minimize price impact.[1164] In contrast, broker-dealers must attempt to execute a marketable held order immediately; these orders better suit retail investors because retail orders typically have much lower price impact, which reduces the need for discretion in order handling.[1165] Brokers-dealers

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1161 See MDI Adopting Release, supra note 10, at 18772-78.

1162 Based on information from broker-dealers' Q2 2023 FOCUS Report Form X-17A-5 Schedule I. This includes both carrying broker-dealers, who maintain custody of customer funds and securities, and introducing broker-dealers, who accept customer orders and introduce their customers to a carrying broker-dealer that will hold the customers' securities and cash. In addition, the Commission acknowledges that the total number of broker-dealers is likely to increase as a result of the recent Dealer Adopting Release. The Dealer Adopting Release adopted new rules to further define the phrase "as a part of a regular business" as used in the statutory definitions of "dealer" and "government securities dealer." The Dealer Adopting Release estimated that up to 43 entities may be required to register with the Commission as a dealer or government securities dealer, which would increase the total number of broker-dealers affected by the amendments.

1163 Customer accounts are identified in CAT as accounts belonging to either the "Institutional Customer" account type, defined as accounts that meet the definition in FINRA Rule 4512(c), or the "Individual Customer" account holder type, defined as accounts that do not meet the definition of FINRA Rule 4512(c) and are also not a proprietary account.

1164 See Securities Exchange Act Release No. 84528 (Nov. 2, 2018), 83 FR 58338 (Nov. 19, 2018) (adopting new order handling disclosure requirements) at nn.59-60 and corresponding text.

1165 FINRA's best execution obligation requires that, "A member must make every effort to execute a marketable customer order that it receives fully and promptly." See FINRA Rule 5310 (Best Execution and serving individual investors often distinguish themselves by the customer service and financial advice they provide and the accessibility and functionality of their trading platforms.

Many broker-dealers that handle customer accounts do not directly access national securities exchanges or ATSs for their orders. They use other broker-dealers to facilitate market access for them through those broker-dealers' order entry systems. The Commission estimates that there are 1,161 broker-dealers with order entry systems that originate orders in NMS stocks in the minimum pricing increments; the amendments to Rule 612 may require changes to these order entry systems.[1166] Of these broker-dealers, an estimated 270 broker-dealers operate smart order routers to facilitate order routing.[1167]

5. Amendments to Rule 605

Several commenters requested the Commission consider interactions between the economic effects of these proposed amendments and the proposed amendments to Rule 605.[1168] The amendments to Rule 605 were not included as part of the baseline in the Proposing Release because they were not adopted at that time. The Commission amended Rule 605 on March 6, 2024,[1169] and the requirements of that rule are part of the baseline considered here. With certain exceptions, the amendments to Rule 605 have a compliance date of Dec. 14, 2025,[1170] which is

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Interpositioning), Supplementary Material para. .01, available at https://www.finra.org/rules-guidance/rulebooks/finra-rules/5310 (accessed Jun. 18, 2024).

1166 See infra note 1656.

1167 See infra note 1660.

1168 See, e.g., SIFMA Letter II; Virtu Letter II; Citadel Letter I; Equity Market Structure Citadel Letter; Citadel Letter II. See also Rule 605 Proposal, supra note 117.

1169 See Rule 605 Amendments, supra note 10.

1170 See supra note 1044. As an exception, after odd-lot order information sufficient to calculate best available displayed price is made available pursuant to an effective NMS plan, market centers, brokers and dealers will have six months to begin including price improvement statistics relative to best available displayed price in their Rule 605 reports. See Rule 605 Amendments, supra note 10, at 26497.

after the compliance dates of the amendments made by this adopting release. The following discussion reflects the Commission's assessment of the anticipated economic effects of the amendments to Rule 605 described in the Rule 605 Amendments as they relate to the baseline for the adoption of these amendments. Specific interactions between the expected economic effects of the amendments to Rule 605 and those of rules adopted herein will be discussed in detail in a later section.[1171]

Rule 605 requires disclosures for order executions in NMS stocks.[1172] The Rule 605 amendments modified reporting requirements in several ways. First, the amendments expanded the scope of reporting entities subject to the rule to include larger-broker-dealers[1173] in addition to market centers.[1174] The amendments also enhanced the accessibility of the reported execution quality statistics by requiring all reporting entities to make a summary report available.[1175]

The Rule 605 Amendments also included amendments to the information required to be reported under Rule 605, some of which are expected to be relevant to the amendments to this Rule. First, the amendments to Rule 605 added requirements related to the reporting of price improvement statistics relative to the best available displayed price, which incorporates information about the best priced odd-lot orders, in addition to the preexisting requirement to

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1171 See infra section VII.E.6.a; see also supra section II.

1172 17 CFR 242.605.

1173 The term "larger broker-dealer" refers to a broker-dealer that meets or exceeds the "customer account threshold," as defined in Rule 605(a)(7) as broker-dealers that carry or introduce orders on behalf of 100,000 or more customer accounts through which transactions are affected for the purchase sale of NMS stocks. See Rule 605 Amendments, supra note 10, at 26428 n.61; 17 CFR 242.605(a)(7).

1174 Regulation NMS defines the term "market center" to mean any exchange market maker, OTC market maker, ATS, national securities exchange, or national securities association. See 17 CFR 242.600(b)(55).

1175 See Rule 605 Amendments, supra note 10, at 26428.

report price improvement statistics relative to the NBBO.[1176] The Rule 605 Amendments acknowledged that, while under the MDI Rules odd-lot information will include pricing information about odd-lots priced better than the NBBO,[1177] the MDI Rules have been approved but not yet implemented, and thus this information is not yet available. Therefore, the Commission stated that Rule 605's price improvement statistics that are relative to the best available displayed price will not be required to be reported until six months after odd-lot order information needed to calculate the best available displayed price is made available pursuant to an effective national market system plan.[1178]

Second, the amendments to Rule 605 require the separate reporting of non-marketable limit orders that are priced at the midpoint of the NBBO or better ("midpoint-or-better NMLOs"), and additionally requires the reporting of information about the price improvement offered to these orders.[1179] An analysis by the Commission in the Rule 605 Amendments indicates that a high percentage of midpoint-or-better NMLO share volume is submitted with

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[1176] See 17 CFR 242.600(b)(14) (defining the "best available displayed price" as, with respect to an order to buy, the lower of: the national best offer at the time of order receipt or the price of the best odd-lot order to sell at the time of order receipt as disseminated pursuant to an effective transaction reporting plan or effective national market system plan; and, with respect to an order to sell, the higher of: the national best bid at the time of order receipt or the price of the best odd-lot order to buy at the time of order receipt as disseminated pursuant to an effective transaction reporting plan or effective national market system plan. With respect to a midpoint-or-better limit order, the best available displayed price shall be determined at the time such order becomes executable rather than the time of order receipt) and 17 CFR 242.605(a)(1)(ii)(M) – (Q).

[1177] See MDI Adopting Release, supra note 10, at 18753.

[1178] In the Rule 605 Amendments, the Commission acknowledged that it was still considering the proposed changes discussed in the Proposing Release and adopted herein, including accelerating the implementation of the round lot and odd-lot information definitions contained in the MDI Release and amending the definition of odd-lot information to include a new data element for the best available odd-lot orders available in the market. In the Rule 605 Amendments the Commission stated that, if it determined to adopt an amendment to the definition of odd-lot information to include a data element that identifies the best odd-lot orders available in the market, reporting entities would be required to use such information to determine the best available odd-lot price. See Rule 605 Amendments, supra note 10, at 26428 n.719.

[1179] See 17 CFR 242.600(b)(57) (defining "midpoint-or-better orders") and 17 CFR 242.605(a)(1)(ii).

IOC designations as compared to other NMLOs, confirming that many of these orders are submitted by traders with the intention of executing immediately against hidden or odd-lot inside-the-quote liquidity, and that these orders tend to have different execution characteristics than other types of NMLOs.[1180] Therefore, the Commission stated that market participants will benefit from an increase in transparency by the separate reporting of these orders, along with the required reporting of certain execution quality statistics that measure the cost of executing immediately, such as effective spreads.[1181]

Third, the amendments to Rule 605 require the reporting of information regarding the extent to which orders received an execution at prices at or better than the quote for share quantities greater than the displayed size at the quote, i.e., "size improvement." This information includes (1) a benchmark metric that measures the displayed size at the time of order receipt, which can then be compared to the number of submitted shares to determine the extent to which a trading venue handled orders that outsized available displayed depth,[1182] and (2) for orders that outsized available displayed depth, the number of shares that received size improvement.[1183]

The amendments to Rule 605 also modified the definition of order size categories from order size categories based on numbers of shares, with orders less than 100 shares excluded, to order size categories based on a notional dollar value range, along with an indication that the

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1180 See Rule 605 Amendments, supra note 10, at 26528.

1181 See Rule 605 Amendments, supra note 10, at 26556-26557, 26568.

1182 See CFR 242.600(b)(72) (defining the "order size benchmark") and 17 CFR 242.605(a)(1)(ii)(R).

1183 See 17 CFR 242.605(a)(1)(ii)(S), requiring the reporting of "the sum of, for each execution of a covered order, the greater of: the total number of shares executed with price improvement plus the total number of shares executed at the quote minus the order size benchmark, or zero." The "total number of shares executed with price improvement plus the total number of shares executed at the quote minus the order size benchmark" ("net size improvement") will only be a strictly positive number for those orders that are both eligible to receive size improvement and actually receive size improvement, and thus is equivalent to a measure of shares that are eligible to and that received size improvement. See Rule 605 Amendments, supra note 10, at 26428 n.1544.

category reflects orders that were for an odd-lot, a round lot, or less than a share.[1184] The Commission stated in the Rule 605 Amendments that one of the benefits of this change is to ensure that round lots for stocks with prices greater than $250 are not excluded from Rule 605 reports following the change in round lot definition under the MDI Rules.[1185]

In the Rule 605 Amendments, the Commission stated that the amendments to Rule 605 will promote increased transparency of order execution quality, particularly for larger broker-dealers who were not required to disclose execution quality information under preexisting Rule 605, but also for market centers, whose execution quality information will be more relevant and easier to access because of improvements to existing Rule 605 disclosure requirements.[1186] The Commission stated in the Rule 605 Amendments that this increase in transparency is expected to increase the extent to which market centers and broker-dealers compete on the basis of execution quality, as well as improvements in execution quality.[1187] The Commission also stated that the amendments to Rule 605 will result in initial and ongoing compliance costs, the majority of which will be related to expanding the scope of reporting entities to include larger broker-dealers, but a significant portion of which will result from the need for market centers to update their systems to process and store the data necessary to prepare the amended reports.[1188]

D. Benefits, Costs, and Other Economic Effects

The Commission expects the adopted minimum quoting increment will alleviate tick constraints and better allow prices to be determined by the forces of supply and demand,

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1184 See 17 CFR 242.600(b)(18).

1185 See Rule 605 Amendments, supra note 10, at 26523.

1186 Id. at 26543.

1187 See Rule 605 Amendments, supra note 10, at 26543-26544.

1188 Id. at 26579-26580.

lowering transaction costs for investors. A lower access fee cap will further reduce the transaction costs of liquidity demanders in the predominant maker-taker structure. Making fees and rebates determinable at the time of trade may enhance broker-dealer order routing by helping mitigate a potential conflict of interest and providing clarity in terms of all in execution costs. Accelerating the inclusion of odd-lot information into the exclusive SIPs, accelerating the implementation of the round lot definitions, and amending the definition of odd-lot information to include the best odd-lot order, will accelerate some of the benefits of the MDI Rules, and could also lead to better order execution by enhancing benchmarking. The amendments will also impose compliance costs on various market participants.

The Commission continually monitors the national market system and the operation of federal securities laws. As discussed above, the national market system continually changes and the Commission, consistent with its oversight of the national market system, will monitor the impact of the adopted rules. With regard to the amendments adopted herein, by May 2029 (three years from the last implementation date), Commission staff will review and study the effects of the amendments in the national market system. Such a review and study might include, but would not be limited to, an investigation of: (i) general market quality and trading activity in reaction to the implementation of the variable tick size, (ii) the reaction of quoted spreads to the implementation of the amended access fee cap, and (iii) changes to where market participants direct order flow, e.g., to exchange versus off-exchange venues, following the implementation of the amendments.

In studying the effect on market quality, a number of different metrics could be examined including quoted, realized, and effective spreads; cumulative depth from the midpoint across multiple price levels; and the cost of a round-trip trade for various trade sizes.[1189] In such analysis, improvements in market quality for stocks affected by Rule 612 would correspond to reduced spreads (adjusting for fees or rebates) or a reduced cost of a round-trip trade.[1190] To isolate the effect of Rule 610, the analysis might focus on those stocks not directly affected by Rule 612. Such analysis might focus on the effect of Rule 610 on quoted spreads (e.g., to examine how the quoted spread adjusts in response to changes in fees and rebates), on whether Rule 610 leads to any change in effective spread off-exchange (due to adjustments to on-exchange quotes), and on any migration of liquidity off-exchange.

1. Modification of Rule 612 to Create a Half-Penny Tick

The Commission is adopting amendments to Rule 612 that introduce one minimum pricing increment that is less than $0.01, i.e., $0.005, for quotes and orders priced $1.00 or more for NMS stocks that have a TWAQS of $0.015 or less during the evaluation period.[1191] Hence, the amendments to Rule 612 will create a smaller tick size for some NMS stocks.

The Commission expects that, on average, market quality will improve for the stocks receiving the smaller tick size. A smaller tick has two competing effects on market quality. First, a smaller tick leads to pricing that more effectively balances liquidity supply and demand, limiting distortions, and thus lowering transaction costs. Second, a smaller tick fragments liquidity in the order book into more price levels, which can increase complexity associated with implementing trades, and increases the incidence of pennying[1192] – effects that can harm

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[1189] *Compare* Table 8.

[1190] One possible study design could focus on stocks close to the TWAQS threshold. Comparing stocks with similar levels of liquidity ex ante would better isolate the effect of the smaller tick size on market quality.

[1191] See infra section III.C

[1192] See supra section I.A.1 and note 994 for the definition and discussion of pennying.

liquidity. A smaller tick can also increase message traffic which can be costly for market participants. The amendments will not change the tick for NMS stocks priced below $1.00, nor for stocks with time-weighted average quoted spread always greater than $0.015 during an Evaluation Period and thus the tick size amendments are expected to have minimal if any effect on the trading environment for these stocks.

a. Estimates of Percent of Trading Volume and Number of NMS Stocks Affected

As discussed in section VII.C.1, prior to these amendments, the tick size for orders in NMS stocks priced equal to or greater than $1.00 was $0.01, and the tick size for orders in NMS stocks priced less than $1.00 was and remains $0.0001.[1193] The amendments assign each NMS stock to one of two tick sizes: $0.005 or $0.01, depending on the stock's time-weighted average quoted spread during an Evaluation Period (specifically, assigning $0.005 for stocks with a TWAQS of $0.015 or less).[1194] Table 7 presents estimates of the amount of share and dollar trading volume that would have been associated with the two tick sizes, as well as the sub $1.00 tick size, based on 2023 trading volumes. It also presents estimates based on the Proposal which would have reduced tick sizes for stocks with TWAQS of $0.040 or less.

Table 7: Estimated Number of Stocks and Trading Volume in Each Tick Size Group[a]

Average Quoted Spread	Tick	Number of Stocks	Estimated % Share Volume	Estimated % Dollar Volume
All Stocks				
Spread <= $0.015	$0.005	1,788	66.2%	42.9%
$0.015 < Spread	$0.01	9,047	33.8%	57.1%

1193 See supra section VII.C.1.a.

1194 See supra section III.C for further discussion.